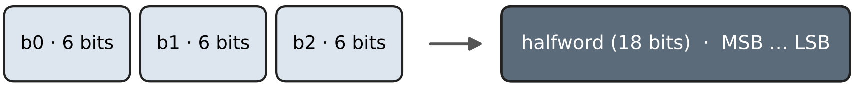
SEISF / DLT cassette layout (Viking Lander 2)

Scrambled instrument buffer as cassettes (UTIG vkg.1–46)

(A) UTIG cassette file (one subgroup)

Header bytes 0–1: tape_id=5 · 2–7: ASCII “DLT…” · 8–9: file no. · 10–11: record_length

(B) Body packing: sequential 18-bit halfwords

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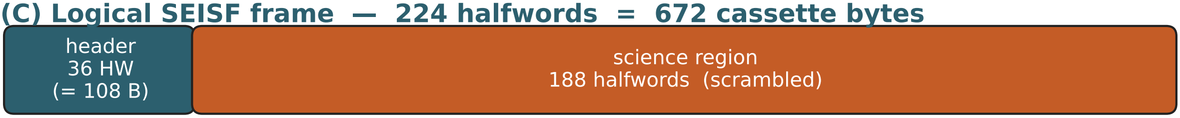
body[i : i+3]  →  h = (b0<<12) | (b1<<6) | b2
```

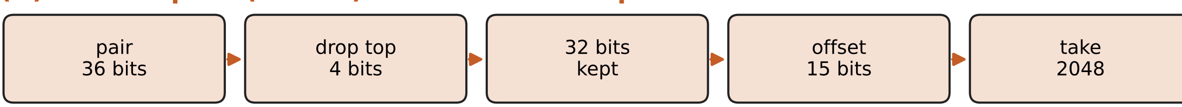
(C) Logical SEISF frame — 224 halfwords = 672 cassette bytes



Frame pitch 224 HW · first base on gold reel ≈ halfword index 170 of record body

Shared eng. header fields (year, DOY, …) match VUS 108-byte prefix after unpack.

(D) Science path (Path D) — from halfword pairs to 2048-bit buffer



Unscramble with PD7400072 page order using Q = matv (MAP probe on first data pair).

Four 512-bit pages → sequential 2048-bit science (same region as VUS bits 648…2695).

(E) 2048-bit science after unscramble (4 × 512-bit pages)

Record boundary note

A frame header near the end of a 10752 B record may need science halfwords from the next physical record. Decoder performs halfword lookahead across record edges.